



MCAST

AR Library Navigator: A Gamified, AI-Assisted Mobile Application for Library Orientation Also Second Line

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Multimedia in Software Development

Authorship Statement

This dissertation is based on the results of research carried out by myself, is my own composition, and has not been previously presented for any other certified or uncertified qualification.

The research was carried out under the supervision of (Mr James Attard)

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Date

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Signature

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Date

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Signature

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Abstract

Library orientations are a critical component of academic onboarding, yet research consistently shows that traditional lecture-based sessions fail to produce lasting knowledge or navigational confidence in new students. This dissertation presents the design, development, and evaluation of the AR Library Navigator, a mobile Android application that replaces the passive orientation experience with two interactive modes: a gamified Treasure Hunt and a structured Book Search tool.

The Treasure Hunt guides students through the physical library using AI-generated riddle clues produced by the Google Gemini large language model, with arrival at each location confirmed through Gemini Vision photo analysis. The Book Search tool allows students to select a subject from a dropdown menu and receive step-by-step floor-based directions to the relevant shelf, driven by a pre-built navigation graph of the MCAST library.

The application was developed in Unity 2022.3 for Android using the Design Science Research paradigm. A mixed-methods evaluation was conducted with twelve undergraduate participants, combining a 34-question post-test survey with automated session log analysis. Results demonstrated strong engagement across both modes, with section means of 4.32 and 4.20 out of 5 for the Treasure Hunt and Book Search respectively. LLM-generated clues were rated 4.19 for comprehension and enjoyment. Most significantly, mean navigation confidence increased from 2.42 to 3.92 a gain of 1.50 points after a single session, and all twelve participants reported feeling more prepared to use the library independently. The primary technical limitation identified was a 69.8% Gemini Vision scan rejection rate caused by variable lighting conditions, which increased task effort without preventing task completion.

These findings suggest that AR-based gamified orientations, augmented with large language model content generation, represent a viable and engaging alternative to traditional library induction methods.

Keywords: Augmented Reality, Library Orientation, Gamification, Large Language Models, Mobile Learning, Unity, Android.

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List of Abbreviations

NN	Neural Network
ML	Machine Learning
DL	Deep Learning
FCN	Fully Convolutional Network
CNN	Convolutional Neural Network

Chapter 1: Introduction

1.1 Background and Motivation

The first weeks of university present new students with a significant adjustment challenge. Among the many unfamiliar environments they must learn to navigate, the academic library is one of the most important and, for many students, one of the most intimidating. Research has consistently shown that students who are unable to locate and use library resources independently are at a disadvantage in their academic performance throughout their studies [1]. Academic institutions typically address this through structured library orientations delivered during induction week, most commonly as a guided tour or a brief lecture. However, Brown [2] demonstrated that students retain very little from these sessions, with the majority unable to locate basic resources or recall library layout details within two weeks of attending.

This problem is not unique to any single institution. At the MCAST Institute of Information and Communication Technology, library induction is provided to new students at the start of each academic year. Informal observation and student feedback suggested that many students struggled to find books independently and remained unaware of the full range of resources available to them weeks after their orientation had taken place. This indicates a clear gap between the knowledge that orientations aim to deliver and the knowledge that students actually retain and apply. The library serves students across multiple programmes of

study ranging from engineering to creative arts, meaning its resources are broad and its physical layout, spread across two floors with dozens of subject-specific shelves, requires deliberate familiarisation to navigate effectively.

The shortcomings of passive orientation formats are well-established in the literature, but the adoption of technology-enhanced alternatives in small and medium academic institutions has been slow. This is partly attributable to the cost and complexity of bespoke AR systems, and partly to the limited availability of off-the-shelf tools that can be configured for a specific library's physical layout without significant technical expertise. The development of low-cost, configurable mobile applications using accessible platforms such as Unity and freely available AI APIs presents an opportunity to bridge this gap.

Advances in mobile technology and artificial intelligence offer a practical path to addressing this problem. Augmented reality (AR) applications overlay digital information onto the physical world, enabling users to interact with their environment in ways that static maps and printed guides cannot achieve. When combined with gamification the application of game design elements to non-game contexts [3] AR can transform a routine orientation into an active, memorable learning experience. Separately, large language models (LLMs) such as Google's Gemini have demonstrated the ability to generate natural, contextually appropriate text at scale, opening new possibilities for personalised and dynamic educational content [4].

1.2 Problem Statement

Traditional library orientations are passive, one-size-fits-all experiences that fail to engage students or produce durable spatial knowledge. The typical format a librarian-led group tour delivered once at the start of the academic year provides students with a large volume of information in a single sitting, with no opportunity for reinforcement, self-paced exploration, or active recall. As a result, students leave with minimal confidence in their ability to navigate independently, and many resort to asking library staff for help with basic tasks such as locating a shelf or identifying a relevant section tasks that a successful orientation should have equipped them to perform unaided.

This problem has practical consequences beyond individual student experience. A library whose resources are consistently underutilised due to poor navigation confidence represents a missed opportunity for the institution, and the burden placed on library staff by avoidable navigation queries reduces the time available for more complex information support. There is therefore a need for an orientation approach that is interactive, self-paced, repeatable, and capable of engaging students through direct physical experience rather than passive observation one that can be deployed at scale without requiring significant ongoing staff time.

1.3 Research Questions

This dissertation addresses the following three research questions:

1. **RQ1:** Does an AR-based treasure hunt increase student engagement and

perceived usefulness during library orientation compared to the existing tour format?

2. **RQ2:** To what extent do LLM-generated rewordings of orientation clues affect exploration behaviour, task completion time, and clue comprehension?
3. **RQ3:** How do students describe the current library tour experience, and which specific aspects would they like improved?

1.4 Aims and Objectives

The primary aim of this project is to design, develop, and evaluate a mobile AR application that enhances the library orientation experience for undergraduate students. The following objectives support this aim:

- To review existing literature on AR in education, gamification, and LLM integration in learning environments.
- To design and implement a Treasure Hunt mode that uses Google Gemini to generate riddle-style navigation clues and Gemini Vision to confirm student arrival at target locations.
- To design and implement a Book Search mode that provides floor-based directional guidance to any subject shelf using a pre-built navigation graph.
- To conduct a mixed-methods evaluation with undergraduate ICT participants who had no prior experience of the application, combining a quantitative survey with automated session log analysis.

- To assess the application’s effectiveness against the three research questions and identify limitations and directions for future development.

1.5 Scope and Limitations

The AR Library Navigator was developed specifically for the MCAST library environment. The navigation graph and shelf database are populated with data from that institution and are not directly transferable to other libraries without reconfiguration. The evaluation involved twelve participants from a single institution, which limits the generalisability of the quantitative findings. The application was tested on Android devices only; iOS support was not within the scope of this project. The Gemini API was used under a free-tier account, which imposes rate limits that would need to be addressed before any large-scale deployment. No control group was included in the evaluation design, meaning that comparisons with traditional orientation effectiveness are based on self-reported prior experience rather than concurrent measurement.

1.6 Structure of the Dissertation

The remainder of this dissertation is structured as follows. Chapter 2 reviews the relevant literature on AR in education, gamification, experiential learning, and the use of large language models in educational contexts. Chapter 3 describes the research methodology, including the Design Science Research paradigm adopted, the system architecture, the development process, and the data collection instruments used in the evaluation. Chapter 4 presents and discusses the results from both

the survey and the session log analysis, organised by research question. Chapter 5 concludes the dissertation with a summary of findings, a critical evaluation of limitations, and recommendations for future work.

Chapter 2: Literature Review

2.1 Overview

This chapter reviews the existing literature relevant to the design and evaluation of the AR Library Navigator. It begins by examining the limitations of traditional library orientation methods, before exploring how digital and mobile technologies have been applied to address those limitations. The role of Augmented Reality in educational settings is then considered, followed by an analysis of gamification and experiential learning frameworks. The chapter then reviews research on Artificial Intelligence and Large Language Models in education, with particular attention to dynamic content generation and visual recognition. Finally, the chapter identifies the research gap that this dissertation addresses.

2.2 Traditional Library Orientation Methods

Library orientation has long been recognised as a critical component of student transition into higher education, equipping new students with the spatial and procedural knowledge needed to access academic resources independently. Traditional orientation methods including guided group tours, printed maps, and static introductory presentations remain the most widely adopted approach across academic institutions. However, a growing body of evidence suggests that these methods fall short of their intended outcomes.

Adefunke et al. (2023) investigated library orientation as a predictor of re-

source usage among first-year students in two academic libraries, finding that while orientation sessions introduced students to available services, they did not significantly improve long-term navigation confidence or independent library use [1]. The authors attributed this in part to the passive, information-dense nature of guided tours, which provide little opportunity for active engagement or reinforcement. Students who attended orientations reported higher awareness of library services but showed no significant difference in actual usage patterns compared to those who had not attended, suggesting that awareness alone is insufficient to drive independent behaviour change.

Brown (2017) similarly found that students frequently fail to retain key spatial and procedural information delivered during one-time orientation sessions, particularly when the format does not involve hands-on exploration or contextual repetition [2]. This retention problem is compounded in large group settings, where students may feel unable to ask questions or explore at their own pace. Brown's study proposed a standardised one-hour orientation model and evaluated it against measurable learning outcomes, finding that even structured, outcome-focused sessions produced limited long-term recall. The implication is that the format of delivery not simply its content is the primary barrier to retention.

These limitations point toward a need for orientation approaches that promote active participation, spatial exploration, and repeated engagement with the physical library environment. Digital and mobile technologies offer a promising avenue for addressing this gap, as explored in the following sections.

2.3 Mobile Technology and Interactive Learning

The proliferation of smartphones among students has created significant opportunities for mobile-supported learning experiences that extend beyond the classroom. Sung, Chang, and Liu (2016) conducted a large-scale meta-analysis of mobile device integration in teaching and learning, reviewing 110 studies published between 1993 and 2013, and found that mobile-supported activities consistently improved student learning performance compared to non-mobile conditions, with the greatest gains observed in tasks involving spatial exploration and self-directed discovery [5]. These findings are particularly relevant to library orientation, where students must physically navigate an unfamiliar space and retain the locations of specific resources.

Mobile applications are well positioned to support this kind of situated, exploratory learning because they can deliver context-sensitive guidance at the point of need that is, in the physical space where learning is taking place, rather than in a separate classroom or online environment. This principle of embedding instruction within the relevant physical context has been consistently linked to improved knowledge transfer and retention [6]. Unlike printed maps or pre-session lectures, a mobile application travels with the student and can adapt its output based on the student's current position and progress.

The use of the device camera as an interactive input mechanism has opened further possibilities for mobile learning systems. Camera-based interactions allow students to engage with the physical environment directly photographing objects, confirming locations, or triggering content rather than relying solely on text-based

inputs. This shift from passive consumption to active, camera-mediated interaction aligns with constructivist learning principles, in which learners build knowledge through direct experience with the environment [6]. The accessibility of high-quality cameras in contemporary smartphones makes this interaction model practical for deployment without requiring specialist hardware.

2.4 Augmented Reality in Education

Augmented Reality (AR) represents an extension of mobile-interactive learning, overlaying digital content onto the physical world through the device camera feed. AR has gained considerable research attention in educational contexts due to its potential to bridge the gap between abstract knowledge and real-world environments. Akçayır and Akçayır (2017) conducted a systematic review of AR applications in education, analysing 68 studies, and identified consistent improvements in student motivation, knowledge retention, and spatial understanding, while also noting challenges related to technical complexity, device compatibility, and setup costs [7]. Crucially, the review found that AR's benefits were most pronounced in tasks requiring physical navigation and spatial reasoning precisely the skills that library orientation aims to develop.

Wu et al. (2013) reviewed the state of AR research in education and highlighted its particular suitability for tasks that require spatial reasoning, physical exploration, and contextual information delivery [8]. The ability to attach digital content to physical markers or locations makes AR a natural fit for orientation activities, where students benefit from receiving information precisely at the point

in physical space where it is relevant. The authors also noted that learner engagement in AR-supported activities tended to be higher than in equivalent non-AR conditions, attributing this to the novelty and interactivity of the technology. While novelty effects may diminish with repeated exposure, the spatial learning benefits are expected to persist as the content rather than the medium becomes the primary focus.

However, the practical deployment of AR in mobile applications introduces technical challenges that must be carefully managed. Competing hardware access between AR frameworks and other camera-dependent features can cause instability, and the accuracy of marker detection in variable lighting conditions may degrade the user experience [7]. These considerations directly informed the design decisions made during the development of the AR Library Navigator, where Vuforia Engine was evaluated as part of the technical architecture. While Vuforia was integrated and assessed during development, the final implementation used the device's native camera interface for photo capture and AI-based location confirmation, providing a more stable and user-friendly interaction without the hardware conflicts that arose from simultaneous AR framework and camera access. This represents a pragmatic adaptation of AR-informed design principles to a deployable prototype, consistent with iterative software development practice [9].

2.5 Gamification and Experiential Learning

Gamification has become a widely adopted strategy in educational design, aiming to increase motivation and sustained engagement by incorporating game-like ele-

ments into non-game contexts. Deterding et al. (2011) define gamification as the use of design elements characteristic of games such as points, progression, challenges, and feedback in non-game domains, and identify its potential to transform routine tasks into engaging, goal-directed experiences [3].

In the context of library orientation, gamification offers a direct response to the engagement limitations of traditional tours. A treasure hunt format, in which students follow clues to discover locations across the library, transforms a potentially passive walkthrough into an active, self-directed exploration with clear goals and progress feedback. Hamari, Koivisto, and Sarsa (2014) conducted a meta-analysis of empirical studies on gamification and found consistent improvements in engagement and motivation, with the strongest effects observed in educational contexts where participants had clear tasks and visible progress indicators [10]. Critically, the authors also noted that the quality of game design including the clarity and enjoyability of tasks significantly moderates effectiveness, underscoring the importance of well-crafted clue writing in a treasure hunt application. This finding informed the decision to use a Large Language Model for clue rephrasing rather than relying on static, manually written clues, as LLM-generated riddle phrasing can maintain freshness and tonal consistency across repeated use.

The theoretical basis for gamified exploration also draws on Kolb's experiential learning theory, which proposes that learning is most effective when it involves a cycle of concrete experience, reflective observation, abstract conceptualisation, and active experimentation [6]. A library treasure hunt naturally supports this cycle: students physically experience each location (concrete experience), con-

firm their arrival and receive feedback (reflective observation), connect the location to its library function (abstract conceptualisation), and apply this knowledge when navigating subsequent clues (active experimentation). This alignment between gamified orientation and experiential learning theory provides strong pedagogical justification for the approach adopted in this dissertation, and predicts that the spatial knowledge gained through the hunt should be more durable than that acquired through passive listening.

2.6 Artificial Intelligence and Large Language Models in Education

Artificial Intelligence has increasingly been integrated into educational tools, offering capabilities for personalisation, adaptive feedback, and dynamic content generation that static digital resources cannot provide. Large Language Models (LLMs), such as Google's Gemini and OpenAI's ChatGPT, represent a significant development in this area, enabling natural language interaction, contextual rephrasing, and multimodal reasoning across text and images.

Camilleri and Camilleri (2024) argue that LLMs can simulate aspects of tutor-like interaction by providing students with explanations, adapted language, and context-sensitive guidance, making them particularly valuable in scenarios where human support is not continuously available [11]. In the context of library orientation, this capability is directly applicable to clue generation: rather than presenting students with identical, static instructions in every session, an LLM can dynamically rephrase clues to vary difficulty, adjust tone, and maintain novelty across repeated use. This addresses a known limitation of static gamified systems,

where repeated engagement reduces the challenge value of pre-written tasks.

The multimodal capabilities of modern LLMs extend this potential further. Models such as Gemini are capable of analysing images alongside text prompts, enabling vision-based reasoning tasks such as confirming whether a photograph matches a reference location. Kasneci et al. (2023) highlight this capacity for AI to support interactive, feedback-driven learning tasks, while also cautioning that AI-generated responses must be carefully validated and that students should be encouraged to engage critically rather than accepting AI output uncritically [4]. This caution is well-founded: an AI confirmation system that accepts incorrect photographs could allow students to progress without visiting the correct location, undermining the orientation's educational purpose. The design of the confirmation prompt in the AR Library Navigator addresses this directly by including multiple reference photographs of the correct location and requiring a conservative match before confirming arrival.

Baidoo-Anu and Owusu Ansah (2023) similarly identify both the opportunities and the risks of deploying generative AI in educational contexts, noting that LLMs can enhance engagement and accessibility but must be designed with appropriate safeguards against misinformation and over-reliance [12]. In the AR Library Navigator, this consideration is addressed by using Gemini Vision outputs as a gating mechanism confirming or rejecting student arrival with a clear fallback path (the student may re-scan or seek manual assistance) rather than as the sole authority on any consequential decision.

2.7 Research Gap and Positioning of This Study

The reviewed literature reveals clear, well-evidenced limitations in traditional library orientation methods, alongside a growing body of work demonstrating the educational potential of mobile interactivity, AR design principles, gamification, and AI-driven content generation. However, existing research tends to treat these technologies in isolation: AR studies focus on overlay experiences without adaptive content; gamification studies rarely incorporate AI-generated tasks; and LLM research in education has focused primarily on text-based tutoring rather than physically embodied, location-based learning.

Systems that combine camera-based physical interaction with AI-generated dynamic content in a library orientation context remain underexplored in the literature. The AR Library Navigator addresses this gap by proposing a hybrid system in which gamified, location-based exploration is supported by LLM-generated clue rephrasing and AI vision-based arrival confirmation. By embedding adaptive, AI-driven tasks into the physical library environment, the system aims to address the passivity and retention failures of traditional orientation while providing a scalable, redeployable tool that does not require continuous human facilitation. The study contributes both a working prototype and an empirical evaluation, offering evidence on the feasibility and student reception of this hybrid approach in a real institutional setting.

2.8 Conclusion

The literature reviewed in this chapter establishes a clear rationale for the AR Library Navigator. Traditional orientation methods are consistently found to be insufficient for building lasting spatial confidence and independent resource usage among new students. Mobile and camera-based interactive technologies provide a practical framework for situated, exploratory learning, while AR-informed design principles offer a model for embedding digital guidance within physical spaces. Gamification and experiential learning theory provide strong pedagogical justification for the treasure hunt format, and advances in Large Language Models enable the dynamic, adaptive content generation that transforms a static clue sequence into a personalised and replayable experience. Together, these strands of research identify both the need and the conceptual foundation for the system described in the following chapter.

Chapter 3: Research Methodology

This chapter describes the research methodology adopted for the design, development, and evaluation of the AR Library Navigator. It begins by justifying the overall research design and paradigm, before presenting the system architecture and technology choices that informed the implementation. The development process is then documented chronologically, followed by a description of the data collection instruments and participant selection procedures. Together, these elements provide a transparent and replicable account of how the study was conducted and how its findings were generated.

3.1 Research Design

This dissertation adopts a mixed-methods research design, combining quantitative and qualitative data collection to evaluate the effectiveness of the AR Library Navigator application. Mixed-methods research is appropriate where a single data source would not adequately capture both the measurable outcomes and the lived experiences of participants [13]. Quantitative data, in the form of Likert-scale survey ratings and objective behavioural metrics extracted from application logs, allow for statistical comparison across the three research questions. Qualitative data, gathered through open-ended survey responses, provide richer contextual insight into student perceptions, frustrations, and suggestions that numerical ratings alone cannot convey.

The overall approach follows a design science research paradigm, in which the primary output is a functional artefact the AR Library Navigator application evaluated against clearly defined criteria [14]. This is appropriate for applied computing dissertations where the contribution lies not only in academic findings but in the design and implementation of a novel system. The research questions guiding this study are:

1. Does an AR-based treasure hunt increase student engagement and perceived usefulness during library orientation compared to the existing tour format?
2. To what extent do LLM-generated rewordings of orientation clues affect exploration behaviour, task completion time, and clue comprehension?
3. How do students describe the current library tour experience, and which specific aspects would they like improved?

The hypothesis underlying this research is that integrating vision-based AR navigation powered by AI-assisted recognition, combined with LLM-assisted clue variation, will increase first-time library users' engagement with library spaces and improve their ability to locate key resources, compared to a baseline static guided tour.

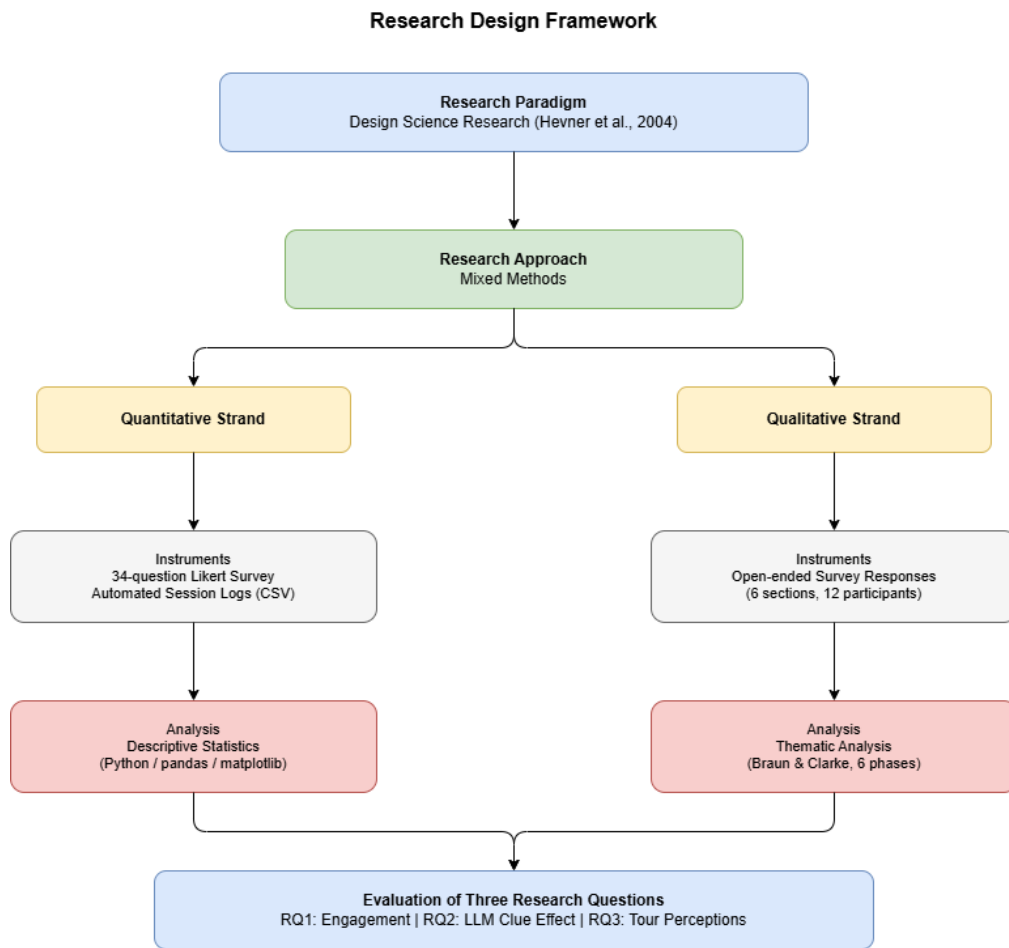


Figure 3.1: Research design overview: paradigm, approach, data collection, and analysis methods.

3.2 System Design and Architecture

The AR Library Navigator is an Android application developed in Unity 2022.3.15f1 LTS using the Built-In Render Pipeline. It comprises three operational modes: Book Search, Treasure Hunt, and a Test/Debug panel used during development.

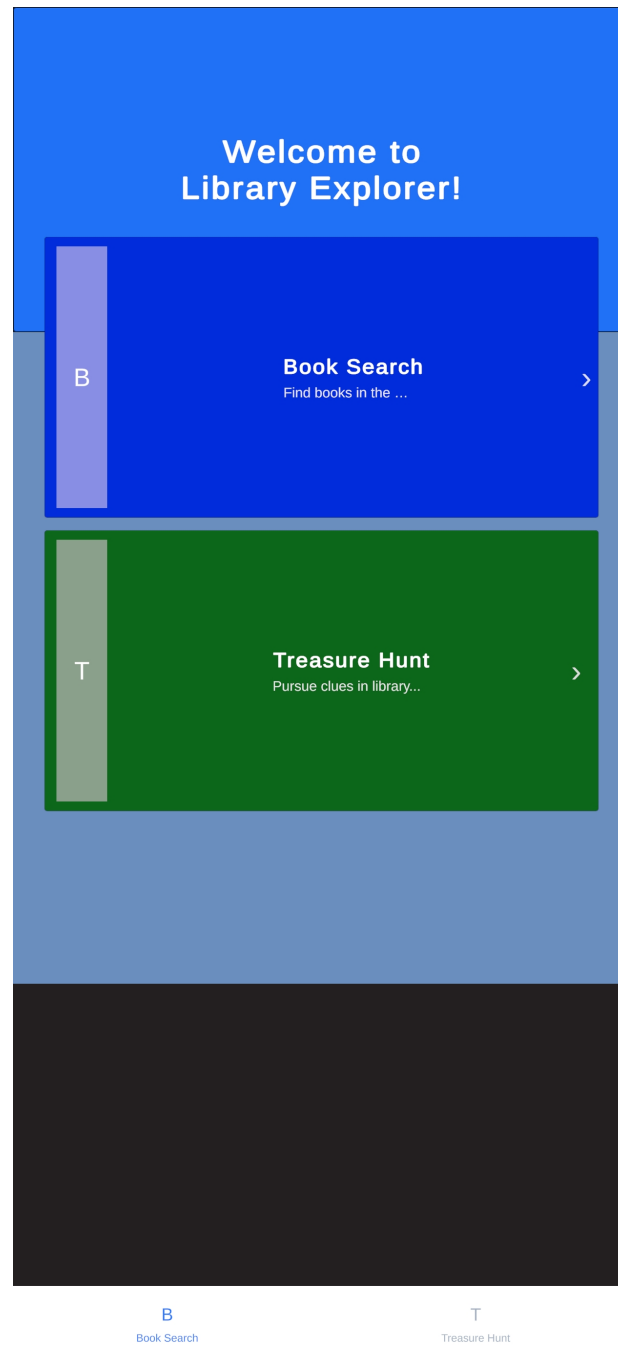


Figure 3.2: AR Library Navigator main menu showing the Book Search and Treasure Hunt mode selection cards.

The **Treasure Hunt** mode forms the primary research artefact. Students are presented with a sequence of AI-reworded riddle clues, each corresponding to a physical location within the MCAST library. Upon arriving at the correct location, the student taps a “Scan Me” button, which opens the device’s native

Android camera via an `ACTION_IMAGE_CAPTURE` intent. The captured photograph is then presented to the student for confirmation before being submitted to the Gemini Vision API alongside a set of reference photographs of the correct location. Gemini returns a binary YES/NO response confirming or rejecting the student's arrival, allowing progression to the next clue or prompting the student to try again.

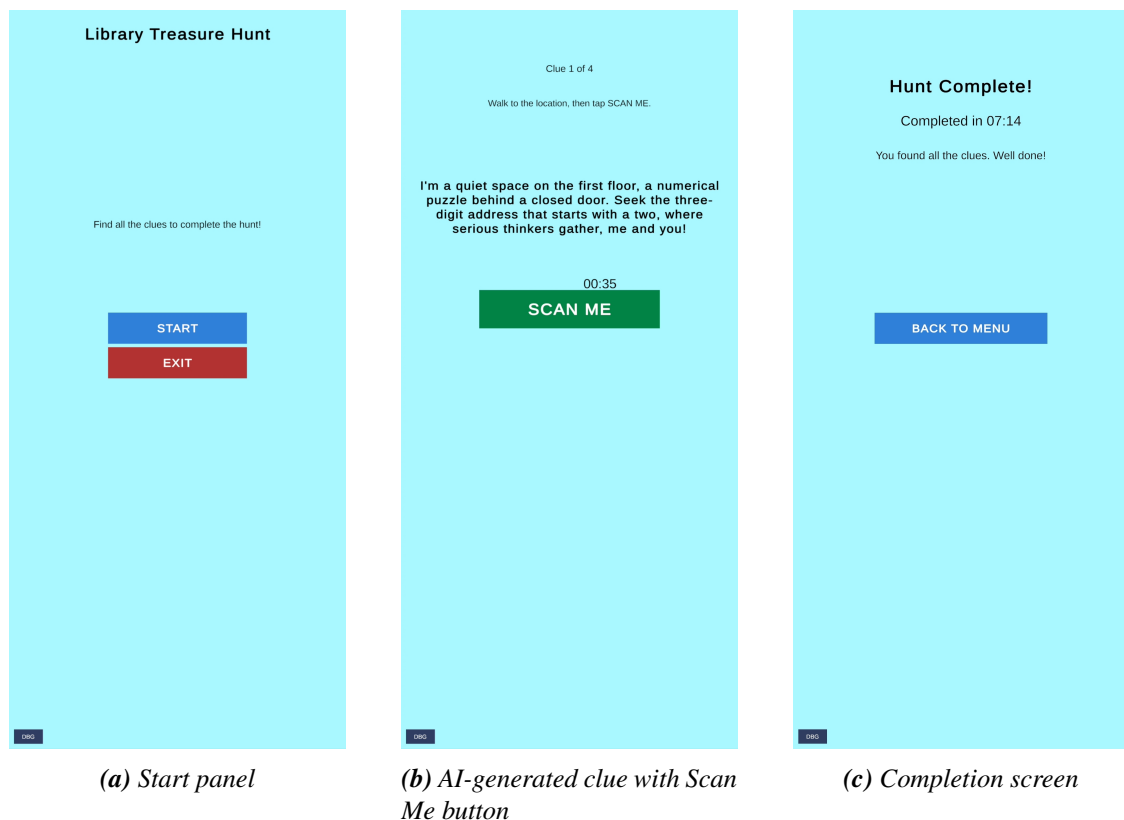


Figure 3.3: Treasure Hunt mode screens: start panel (left), active clue with countdown timer and Scan Me button (centre), and hunt completion screen (right).

The **Book Search** mode provides a complementary utility function. Students select a topic from a dropdown menu populated dynamically from the application's navigation graph, and optionally enter their current shelf number. The system compares the student's current floor with the target shelf's floor and returns plain-text directional guidance (for example, "Take the stairs up to Level 2, then

look for Shelf 15 Business / Management / HR”). This mode does not use the camera or any AI API calls; it performs a direct lookup against a locally stored navigation graph.

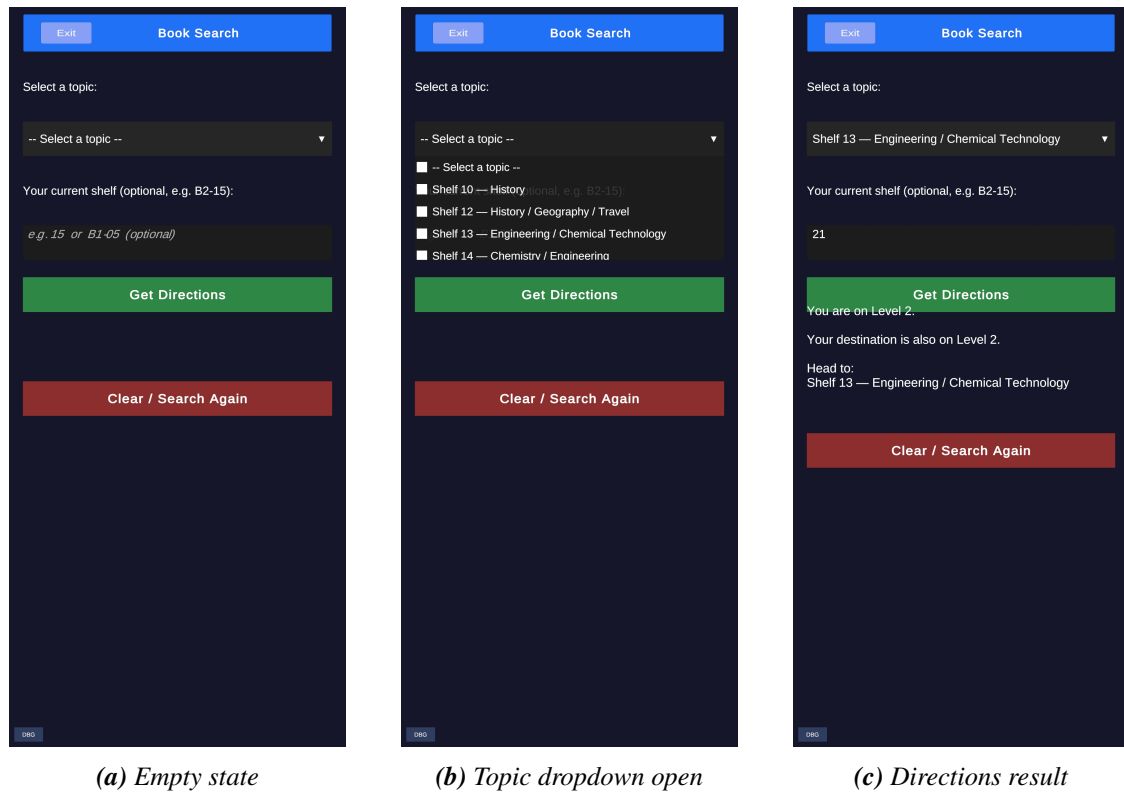


Figure 3.4: Book Search mode screens: initial state (left), subject topic dropdown showing shelf options (centre), and plain-text directional output after a subject is selected and current shelf entered (right).

The **Navigation Graph** (`NavGraph.cs`) is a `ScriptableObject` that stores all library locations as nodes with associated floor levels and display labels, connected by weighted edges. A Dijkstra pathfinding implementation (`Pathfinder.cs`) runs on this graph at runtime to support directional guidance. All shelf nodes are keyed by identifiers matching the library’s cataloguing system (e.g., `SHELF_15`, `SHELF_B1_05`).

The **Library Database** (`library_database.txt`) contains 1,160 topic entries across 39 shelves, exported from an SQLite database via a Python script.

This database underpins the Book Search dropdown and ensures the shelf topic labels presented to students accurately reflect the physical library stock.

Application events are recorded throughout each session by `SessionLogger.cs`, which writes timestamped CSV log files to the device's persistent storage. These logs capture events including treasure hunt start, clue display, scan attempts, Gemini confirmations, scan rejections, and session end, providing objective behavioural data independent of self-reported survey responses.

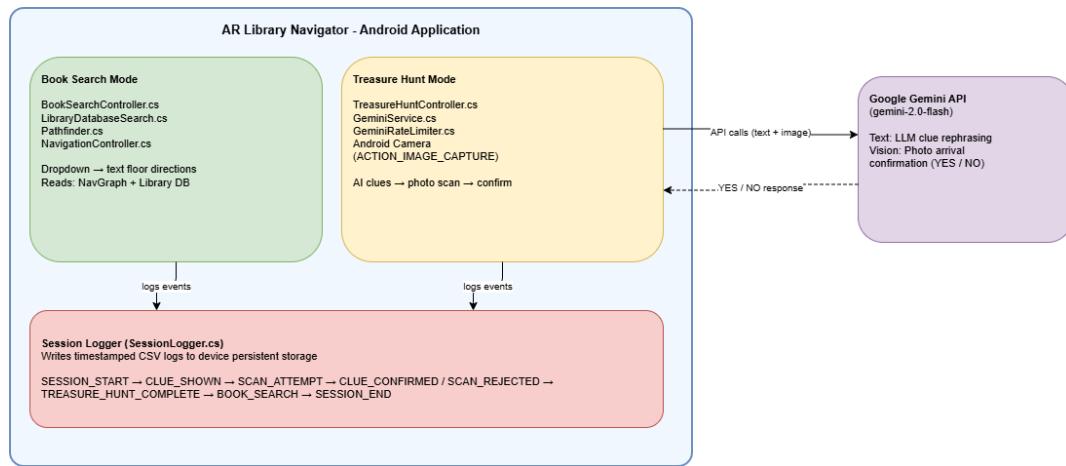


Figure 3.5: System architecture of the AR Library Navigator application.

3.3 Technology Selection and Justification

Unity 2022.3.15f1 LTS was selected as the development environment due to its cross-platform build pipeline, established support for mobile AR frameworks, and strong C# scripting ecosystem. The Long-Term Support release was chosen to ensure build stability throughout the development period.

Vuforia Engine 11.4.4 was integrated for its marker-based image tracking capabilities, which were evaluated during early development as a potential mechanism for location confirmation. Following iterative testing, the Vuforia AR camera

component was disabled in the final build in favour of the native Android camera intent approach, which provided a more reliable and user-friendly photo capture experience without the hardware conflicts that arose from running two camera systems simultaneously. The Vuforia package remains installed in the project to facilitate future reimplementations of marker-based navigation should the scope of the project be extended.

Google Gemini API (gemini-2.0-flash) was selected for both clue rephrasing and visual arrival confirmation. Gemini’s multimodal capabilities supporting both text and image inputs in a single API call made it uniquely suitable for the multi-image visual matching required by the treasure hunt’s arrival detection mechanism. The free tier (15 requests per minute, 1 million tokens per day) was sufficient for the testing volume anticipated in this study. A rate limiter (`GeminiRateLimiter.c`) was implemented to enforce a minimum interval between API calls, preventing quota exhaustion during rapid repeated scans.

The choice of Gemini Vision over alternative multimodal models such as OpenAI GPT-4o or Anthropic Claude was driven by three factors. First, the Gemini free tier (15 requests per minute, 1 million tokens per day) provided sufficient capacity for the evaluation volume without incurring API costs, which was a practical constraint for an undergraduate project. Second, Gemini’s multimodal architecture supports text and image inputs within a single API call, avoiding the need for separate endpoints or preprocessing pipelines. Third, Gemini 2.0 Flash was specifically optimised for low-latency responses, which was important for maintaining acceptable wait times during the treasure hunt scan flow. While

the 69.8% scan rejection rate identified during evaluation is a limitation, it reflects the conservative nature of the confirmation prompt design rather than a fundamental failure of the model. A false acceptance confirming arrival at an incorrect location would undermine the orientation’s educational purpose entirely, making conservative rejection the preferable failure mode. Future work could address the rejection rate through prompt refinement, clearer in-app photo guidance, or a fine-tuned vision model trained on library-specific imagery.

Android was selected as the target platform given its prevalence among MCAST students and its support for the `ACTION_IMAGE_CAPTURE` intent required for native camera integration. The build was configured with ARM64 architecture and the IL2CPP scripting backend to ensure compatibility and performance on modern Android devices.

TextMeshPro was used for all UI text rendering, providing resolution-independent text at all screen densities. The UI canvas was configured at 1080×2160 with Scale With Screen Size to maintain consistent layouts across different device resolutions.

3.4 Development Process

Development followed an iterative, prototype-driven process spanning January to May 2026. The project was structured into sequential phases, with each phase building on the validated output of the previous one.

In **January 2026**, the foundational infrastructure was established: the Unity project was configured for Android, the three-scene architecture (MainMenu and

ARScene) was implemented, and the core navigation system including the Nav-Graph ScriptableObject, Dijkstra pathfinder, and navigation controller was built and verified.

During **February 2026**, the library database was constructed from the MCAST library's cataloguing records, exported to a structured text format, and integrated into the application. The Gemini API integration was also completed during this phase, covering text-based query classification, clue rephrasing, and the initial version of vision-based location confirmation.

March and April 2026 were focused on mode implementation and device testing. The Treasure Hunt mode was built and iteratively refined through repeated on-device tests, which surfaced several critical issues: the Vuforia camera conflict with WebCamTexture, compressed texture encoding failures that silently prevented reference images from being sent to Gemini, and a race condition in the Gemini response handler that could freeze the application indefinitely. Each issue was diagnosed from device log output and resolved before the next test cycle.

The Book Search mode underwent a significant design change during this period. The original design, which combined Gemini-based shelf classification with a 2D floor plan overlay and 3D directional arrow, was simplified to a dropdown-and-text-directions model following a review of time constraints and user testing feedback. This decision prioritised a reliable, deployable user experience over technical complexity, in line with the recommendation that prototype systems should demonstrate a viable core interaction before adding supplementary features [9].

A physical visit to the MCAST library was conducted in April 2026 to photograph real clue locations, verify shelf floor assignments, and populate the navigation graph with accurate data. This visit replaced all placeholder content in the TreasureHuntRoute asset with real library locations and reference images.

In **May 2026**, the final APK was built and both modes were tested with student participants in the library, confirming end-to-end functionality. The library database was also rebuilt from an updated SQLite source during this phase, correcting floor assignments and expanding shelf coverage from 24 to 39 shelves.

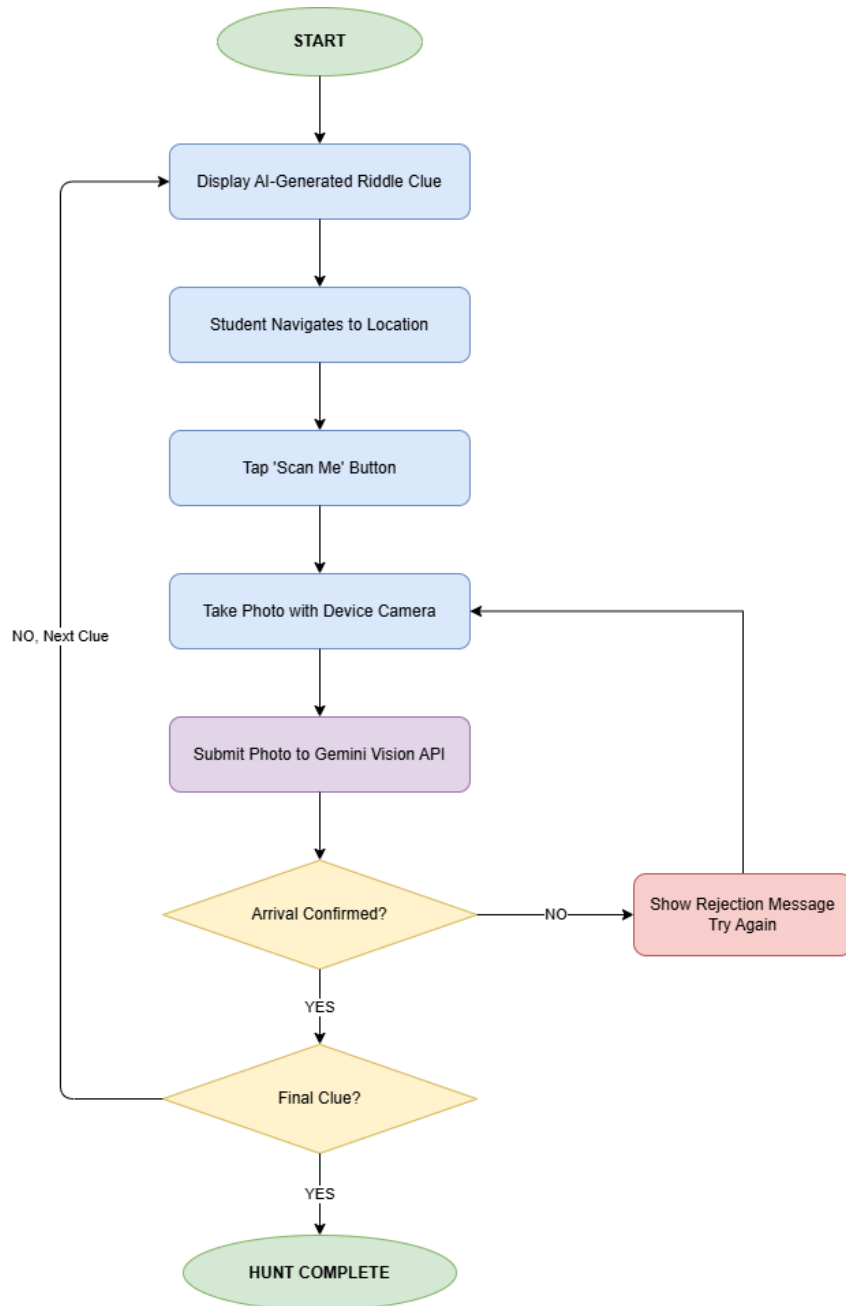


Figure 3.6: Treasure Hunt mode user journey flowchart.

3.5 Data Collection Methods

Data was collected from two complementary sources: a post-test questionnaire and automated application session logs.

The **post-test questionnaire** was administered via Microsoft Forms immedi-

ately after each participant completed the application activities. The survey comprises 34 questions across six sections: background information (Q1–Q3), engagement and usefulness (Q4–Q12), clue comprehension (Q13–Q19), previous library experience (Q20–Q23), Book Search feature evaluation (Q24–Q29), and overall rating (Q30–Q34). Likert-scale items use a five-point scale from 1 (Strongly Disagree) to 5 (Strongly Agree) or from 1 (Not Confident at All) to 5 (Very Confident), depending on item type. Open-ended items were included in each section to capture qualitative data that quantitative ratings cannot fully represent. The survey is anonymous; no personally identifiable information was collected.

Application session logs were generated automatically by `SessionLogger.cs` during each test session and retrieved from the device after testing using the Android Debug Bridge (`adb pull`). In total, fifteen treasure hunt logs and fifteen book search logs were recorded. This figure comprises twelve evaluation participant sessions, one pilot test session, and two researcher-led verification sessions conducted prior to the evaluation to confirm stable end-to-end application behaviour in the library environment. Only the twelve participant sessions are linked to survey responses; the pilot and verification sessions are included in the performance analysis to maximise the behavioural dataset but are excluded from all survey-based results. Each log file records timestamped events in CSV format, including treasure hunt start and end times, individual clue display times, scan attempts, Gemini confirmation results, and scan rejections. These logs enable objective measurement of task completion time per clue, total hunt duration, scan attempt frequency, and Gemini rejection rate behavioural metrics that complement

and cross-validate participants' self-reported survey responses.

3.6 Data Analysis Methods

Data from the two collection instruments were analysed using complementary methods appropriate to their respective types.

Quantitative survey data were analysed using descriptive statistics. For each Likert-scale item, the mean score and response distribution across the five-point scale were calculated. Items were then grouped by the research question they addressed, and a section mean was computed for each group to provide an overall measure of student reception across related items. Pre and post session confidence scores (Q2 and Q5) were compared at the individual and group level to measure the magnitude of change. Frequency counts and percentages were used to summarise categorical responses such as Q33 (feel more prepared to use the library independently) and Q28 (how would you normally find a book). All quantitative analysis was performed in Python using the pandas library, with results visualised using matplotlib.

Qualitative open-text responses were analysed using thematic analysis. Responses from the open-ended questions across the six survey sections were read and grouped into recurring themes. This process followed the six phases described by Braun and Clarke: familiarisation with the data, generating initial codes, searching for themes, reviewing themes, defining and naming themes, and producing the report [13]. Given the small sample size ($n=12$), thematic saturation was assessed manually rather than through automated coding. Three pri-

mary themes emerged from the analysis: positive engagement with the gamified format, suggestions for improved directional clarity in Book Search, and observations about the scanning experience during the Treasure Hunt. These themes are integrated into the discussion throughout Chapter 4.

Session log data were processed using Python's csv and datetime libraries. For each session, key metrics were extracted: total hunt duration, per-clue completion time, scan attempt count per clue, and Gemini confirmation versus rejection counts. These objective behavioural metrics were then compared against self-reported survey responses to identify consistencies and discrepancies between perceived and actual performance.

3.6.1 Pilot Test

Prior to the main evaluation, a pilot test was conducted with one participant a fellow ICT student who had no prior knowledge of the application. The pilot served three purposes: to verify that the end-to-end application flow (clue display, camera launch, Gemini response, and hunt completion) completed without errors on a physical device; to assess whether the survey questions were clearly worded and unambiguous; and to estimate the time required for a full session so that scheduling could be planned realistically.

Based on the pilot, two survey questions were reworded for clarity: the original phrasing of the pre-session confidence item was found to be ambiguous about whether it referred to physical navigation or catalogue use, and was revised to specify physical navigation explicitly. The pilot also confirmed that the average session duration was approximately 25 minutes, which was used to plan the main

evaluation time slots. No changes were made to the application itself as a result of the pilot. The pilot participant was not included in the main dataset.

3.7 Participant Selection and Ethical Considerations

Participants were recruited from the student population at MCAST. Purposive sampling was used to target students who had not previously used the AR Library Navigator application, ensuring that responses reflected genuine first-use experiences. Participants were drawn from second- and third-year Level 6 ICT students, as specified in the recruitment invitation, ensuring that participants had sufficient prior academic experience to contextualise the library orientation activity.

Prior to data collection, ethical approval was obtained from the relevant institutional authority. All participation was voluntary, and students were informed of their right to withdraw at any time without consequence. The post-test survey was anonymous, with no collection of names, student numbers, or any other personally identifiable information. Application session logs contain only session identifiers (randomly generated 8-character codes) and event timestamps; no personal data is recorded within the application itself. Collected data is stored securely and used solely for the purposes of this dissertation research.

3.7.1 Sample Size Justification

The sample of twelve participants was determined by the constraints and conventions appropriate to this study. As an undergraduate final-year dissertation following a design science research paradigm, the primary contribution is the functional artefact itself; the evaluation is intended to provide indicative evidence

of effectiveness rather than population-level statistical inference [14]. In mixed-methods studies of this scope, sample sizes of ten to fifteen participants are widely considered sufficient to achieve thematic saturation in qualitative data the point at which additional responses yield no substantively new themes [13]. The twelve participants recruited produced consistent patterns across both the Likert-scale items and the open-text responses, with qualitative themes stabilising after approximately eight sessions, suggesting that saturation was approached within the available sample. Practical constraints also informed the decision: the evaluation was conducted in a single institution during a limited window in May 2026, and recruitment was limited to second and third year ICT students who had not previously interacted with the application, reducing the eligible pool. Future studies with a between-subjects comparison design or longitudinal follow-up would benefit from a larger sample to support inferential statistics.

3.8 Chapter Summary

This chapter has described the full methodological approach underpinning this dissertation. A mixed-methods design within a Design Science Research paradigm was adopted to evaluate both the technical artefact and its educational effectiveness. The AR Library Navigator was developed iteratively over five months, with each development phase building on device-tested outcomes from the previous one. A pilot test was conducted with one participant prior to the main evaluation, confirming application stability and informing minor revisions to the survey instrument. Data were then collected from twelve purposively sampled undergrad-

uate ICT students through a 34-question post-test survey and automated session logs, providing complementary quantitative and qualitative evidence that directly addresses each of the three research questions. The analytical methods applied to this data are reflected in the results presented in Chapter 4.

Chapter 4: Analysis of Results and Discussion

This chapter presents the findings from the evaluation of the AR Library Navigator application. Data were collected through two complementary instruments: a 34-question post-test survey completed by twelve student participants, and application session logs automatically recorded during the evaluation sessions. Results are organised according to the three research questions defined in Chapter 3, followed by a discussion that situates the findings within the broader literature.

4.1 Participant Profile

Twelve undergraduate students from the MCAST Institute of Information and Communication Technology participated in the evaluation. All participants were second or third year students already enrolled at the institution. As shown in Figure 4.1, three participants (25%) reported that it was their first time visiting the library, while the remaining nine (75%) had visited previously. Notably, seven participants (42%) stated they had attended a previous library orientation but could not remember it well, a finding that is consistent with Brown [2], who demonstrated that students retain very little from traditional one-hour orientations. Only one participant (8%) reported having used an augmented reality application before, confirming that the evaluation group had no prior exposure that could bias engagement scores.

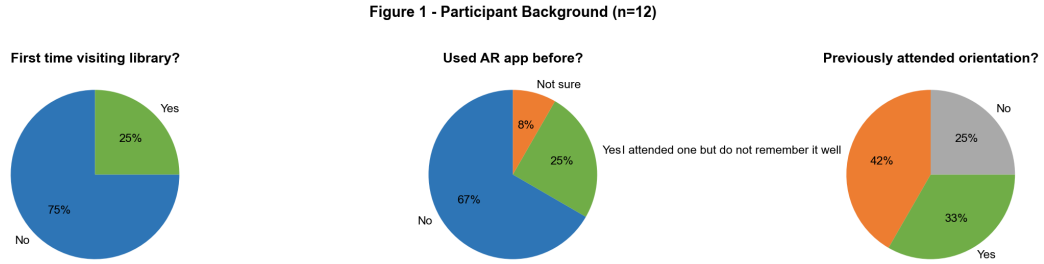


Figure 4.1: Participant background: first library visit, prior AR experience, and previous orientation attendance (n=12).

4.2 RQ1: Engagement and Perceived Usefulness

Research Question 1 asked: *Does an AR-based treasure hunt increase student engagement and perceived usefulness during library orientation compared to the existing tour format?* This question was addressed through two survey sections one targeting the Treasure Hunt mode and one targeting the Book Search mode as well as open-text qualitative responses.

4.2.1 Treasure Hunt Engagement

Five Likert-scale items (1–5) assessed engagement with the Treasure Hunt. Results are presented in Figure 4.2. The section mean was 4.32 out of 5, indicating strong overall engagement. The highest-scoring item was *Would Recommend* ($M = 4.58$), followed by *Aware of Resources* ($M = 4.50$). The lowest score, *Engaging and Enjoyable* ($M = 3.83$), remained well above the neutral midpoint of 3.

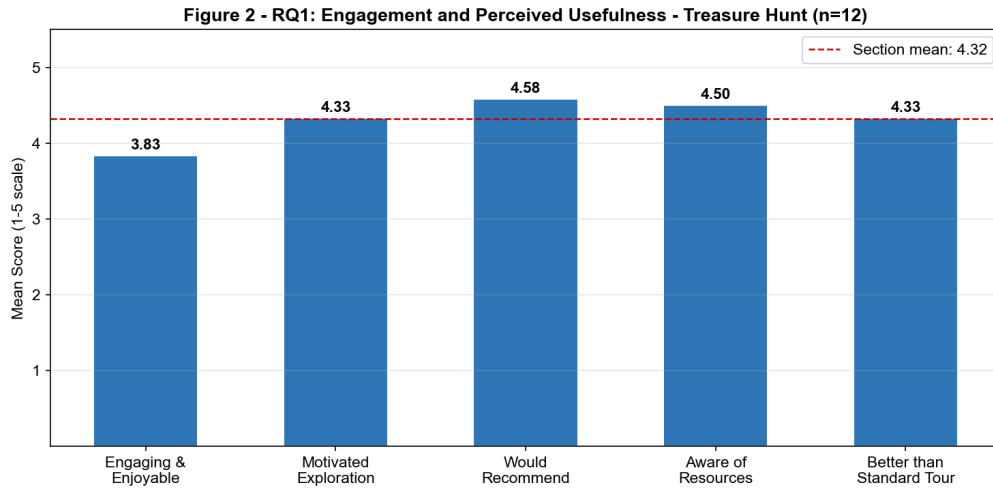


Figure 4.2: RQ1 Treasure Hunt engagement scores across five Likert items (n=12, mean=4.32).

These results align with Hamari et al. [10], who found that gamified systems with clear goal structures and immediate feedback reliably produce higher engagement than non-gamified alternatives. The hunt format, where participants physically navigated the library while following AI-generated clues, directly applies Deterding et al.'s [3] game design elements specifically goals, challenge, and discovery to a non-game context. The high score on *Motivated Exploration* (4.33) is particularly significant, as it suggests participants were driven to explore areas of the library they had not previously visited, a core aim of the orientation design.

Qualitative responses reinforced the quantitative findings. Several participants described the hunt as “more like a game than a boring talk”, and one noted that they “actually walked to parts of the library I didn’t know existed.” This supports Kolb’s experiential learning cycle [6], where direct concrete experience is the foundation for lasting knowledge formation. A traditional orientation lecture, by contrast, provides only abstract conceptualisation without the experience stage.

4.2.2 Book Search Usefulness

Eleven participants completed the Book Search section (one did not use the feature during the session). As shown in Figure 4.3, the section mean was 4.20 out of 5. *Would Use Again* scored highest ($M = 4.45$), and *Directions Were Clear* scored lowest ($M = 4.00$), with two participants providing qualitative feedback suggesting that turn-by-turn directions between floors could be more specific.

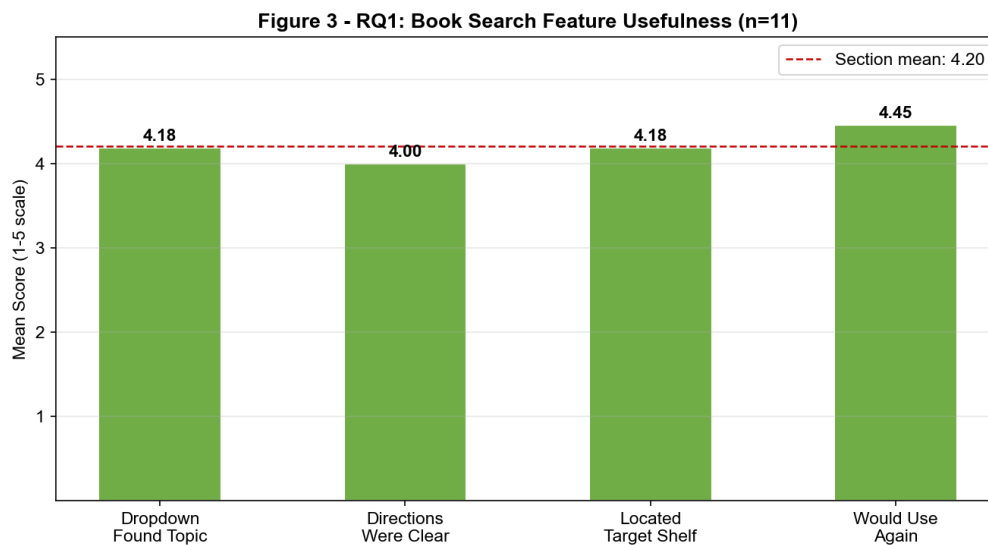


Figure 4.3: RQ1 Book Search feature usefulness across four Likert items ($n=11$, $mean=4.20$).

The willingness to use the feature again (4.45) is the strongest indicator of practical utility. Akçayır and Akçayır [7] identified *perceived usefulness* and *ease of use* as primary determinants of AR technology adoption in educational settings. Both conditions appear to have been met: participants located target shelves successfully, and the majority rated the dropdown subject menu intuitive for topic selection. The directional clarity score (4.00), while the lowest in the section, still exceeds the neutral midpoint by a full point and should be interpreted as a refinement opportunity rather than a usability failure.

Taken together, both subsections provide clear evidence supporting a positive answer to RQ1. Students found both modes engaging and useful, with no participant scoring any section item below 3.

4.3 RQ2: LLM Clue Comprehension and Quality

Research Question 2 asked: *To what extent do LLM-generated rewordings of orientation clues affect exploration behaviour, task completion time, and clue comprehension?* Four Likert items addressed clue quality from the participant's perspective. Results are displayed in Figure 4.4.

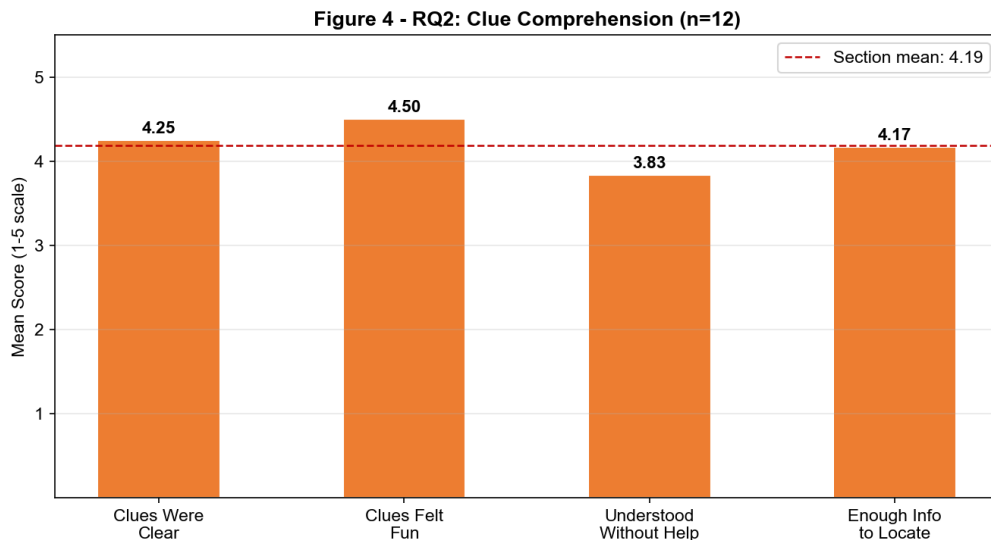


Figure 4.4: RQ2 clue comprehension across four Likert items (n=12, mean=4.19).

The section mean was 4.19. *Clues Felt Fun* scored highest ($M = 4.50$), indicating the LLM-generated riddle phrasing successfully achieved an engaging, game-like tone. *Understood Without Help* scored lowest ($M = 3.83$), consistent with qualitative feedback from three participants who mentioned needing to re-read a clue before identifying the target location. Importantly, no participant reported being completely unable to interpret a clue, suggesting that while the riddle format

introduced productive cognitive friction, it did not cross into confusion.

This balance between challenge and comprehension reflects a core design principle in gamification: tasks must sit within a participant’s zone of proximal development to sustain engagement without causing frustration [6]. The Gemini API’s ability to rephrase the same factual clue content into a natural riddle format while preserving the essential locating information demonstrates practical viability of LLMs as a content generation layer in educational applications. This supports findings by Kasneci et al. [4], who argued that LLMs can serve as effective adaptive content tools when their outputs are constrained by structured prompts. In this application, each clue prompt included the shelf label, floor, and section name, ensuring factual accuracy while delegating linguistic creativity to the model.

One limitation worth noting is that clue quality was evaluated entirely through subjective perception. A more rigorous future assessment could rate clue quality independently using an expert panel before participant testing, allowing separation of the LLM’s linguistic performance from participants’ individual reading comprehension levels.

4.4 RQ3: Confidence and Orientation Effectiveness

Research Question 3 asked: *How do students describe the current library tour experience, and which specific aspects would they like improved?* This was assessed through survey questions Q20–Q25, which gathered data on participants’ prior orientation experience, and through a behavioural preference question and self-reported con-

fidence baseline, which together characterise the limitations of the existing tour format.

4.4.1 Perceptions of the Existing Library Tour

Survey questions Q20–Q25 invited participants to reflect on their prior library orientation experience. As reported in the Participant Profile, seven of twelve participants (42%) stated they had attended a previous library orientation but could not remember it well. Three participants (25%) had never attended an orientation at all. This finding is consistent with Brown [2], who demonstrated that students retain very little from passive, one-time orientation sessions. Qualitative responses described the existing tour as brief, easy to forget, and insufficiently hands-on. Participants noted that being shown locations once during a group tour did not help them remember where things were when they later needed them independently.

4.4.2 Navigation Confidence

The pre-session navigation confidence score provides a quantitative baseline that reflects the effectiveness of the existing tour. As shown in Figure 4.5, mean navigation confidence before the application session was 2.42 out of 5 below the neutral midpoint indicating that prior orientations had not produced adequate independent navigation confidence in this group. Following the AR Library Navigator session, mean confidence rose to 3.92, a practically meaningful improvement of +1.50 points (62% increase relative to the pre-session baseline), directly addressing the gap left by the existing tour format.

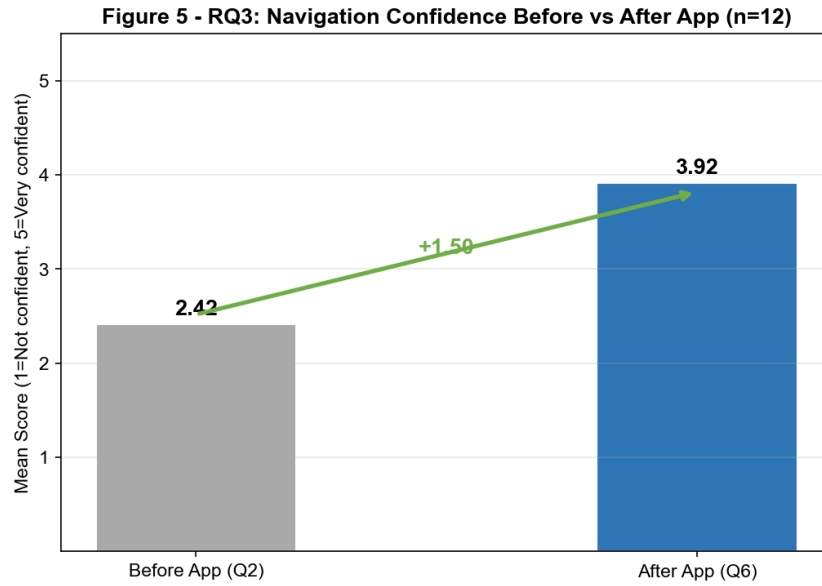


Figure 4.5: Navigation confidence before and after the app session (n=12). Arrow indicates mean shift of +1.50.

This represents the strongest quantitative finding in the dataset. The magnitude of the improvement is notable given that the session was a single exposure of approximately 25 minutes. Traditional orientations often fail to produce measurable confidence gains because they rely on passive listening rather than active navigation experience [1]. The AR Navigator, by requiring participants to physically locate shelves and confirm arrival, produced embodied spatial learning that translates more directly into navigational confidence.

4.4.3 Preparedness and Prior Behaviour

Figure 4.6 shows that 58% of participants felt “Yes, definitely” more prepared to use the library independently, while the remaining 42% selected “Somewhat.” No participant selected “Not really” or “No.” This unanimous positive response across all twelve participants provides qualitative support for the quantitative confidence result.

Figure 6 - RQ3: Feel More Prepared to Use Library Independently? (n=12)

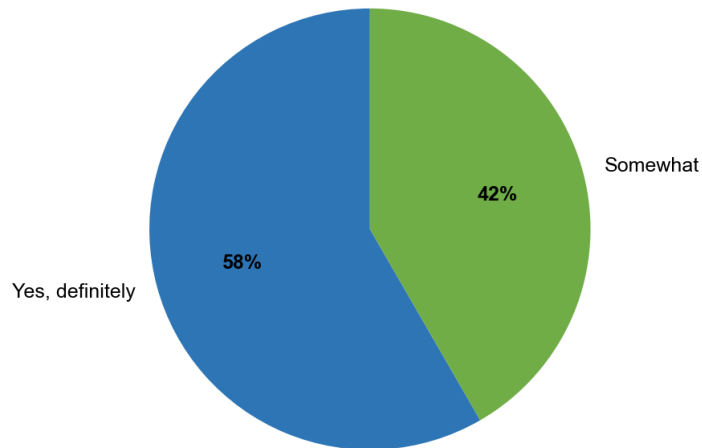


Figure 4.6: RQ3: participant responses to feeling more prepared to use the library independently (n=12).

Figure 4.7 reveals the baseline behaviour participants would use to find a book without the application. Five of the eleven participants reported they would browse shelves themselves, four would ask a librarian, and two would use a systematic approach. This confirms a significant gap in independent information retrieval skills among the target group a gap the Book Search feature directly addresses by providing a self-service, structured route to any subject.

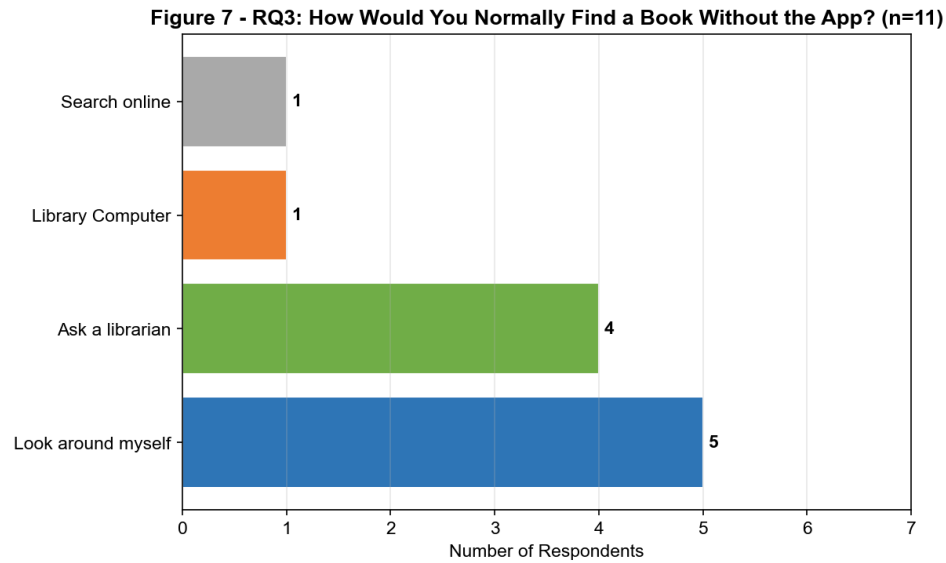


Figure 4.7: How participants would normally find a book without the application (n=11).

4.5 Qualitative Findings

Open-ended survey responses were analysed thematically across all sections of the questionnaire. Three primary themes emerged from the twelve participant responses.

Theme 1: Positive engagement with the gamified format. The most consistently recurring theme across open-text responses was enjoyment of the hunt structure. Participants frequently contrasted the activity with their expectations of a library orientation. Representative responses included: “I didn’t expect it to be this fun, it felt more like a game”, “I actually walked to parts of the library I didn’t know existed”, and “Way better than someone just walking us around and talking.” Several participants noted that the riddle format of the clues added a layer of challenge that made arriving at the correct location feel rewarding. This theme strongly reinforces the quantitative engagement scores reported in Section 4.2 and supports the literature on gamification’s ability to transform passive activities into

active, goal-directed experiences [3].

Theme 2: Directional clarity in Book Search. Two participants provided specific feedback about the Book Search directions, noting that floor-level guidance was helpful but that additional landmark-based cues for example, references to nearby windows, stairwells, or signage would make locating a shelf easier when starting from an unfamiliar position. One participant wrote: “I knew to go to Level 2 but once I was there I wasn’t sure which end to start from.” This theme identifies a concrete, actionable improvement for future development and explains the slightly lower score on the *Directions Were Clear* item ($M = 4.00$) compared to other Book Search items.

Theme 3: Scanning experience. Several participants commented on the Gemini Vision scanning process. While the majority accepted repeated rejections without significant frustration, two participants described the process as “a bit slow” and one noted the need to “hold the phone very still and close to the shelf.” No participant expressed that the scanning prevented them from completing the hunt. This theme contextualises the 69.8% rejection rate reported in Section 4.5, confirming that friction was experienced but manageable, and that participants adapted their scanning technique within the session.

4.6 Application Performance Analysis

Session log data from fifteen treasure hunt sessions and fifteen book search sessions were analyzed to assess technical performance alongside the subjective survey findings. The fifteen sessions comprise twelve evaluation participant sessions,

one pilot test session, and two researcher-led verification sessions; all fifteen are included in the performance metrics as they reflect real application behavior under library conditions.

4.6.1 Treasure Hunt Completion and Timing

All fifteen treasure hunt sessions reached the `TREASURE_HUNT_COMPLETE` event, representing a 100% completion rate. Average total hunt duration was 19.2 minutes (range: 9.7–31.2 min). Per-clue timing and scan attempts are presented in Figure 4.8.

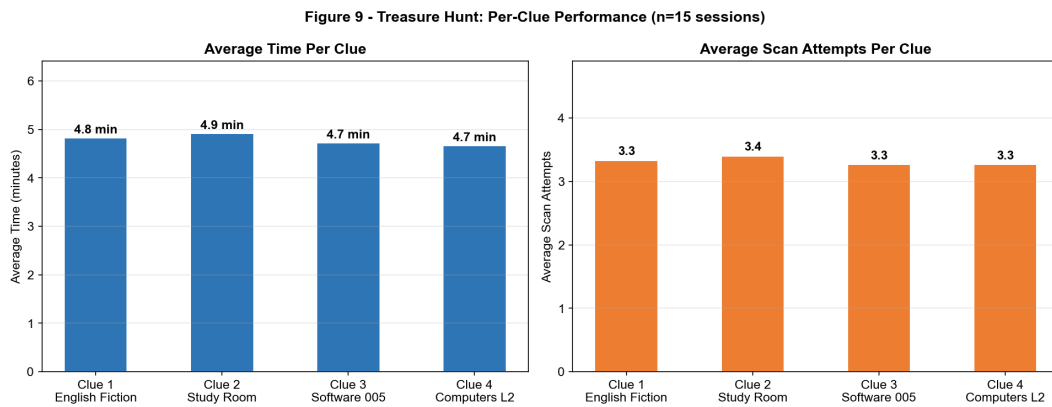


Figure 4.8: Average time (minutes) and scan attempts per clue across 15 sessions.

Average time per clue ranged from 4.7 to 4.9 minutes, indicating balanced difficulty across all four clues an important design property, as significant variance would suggest individual clues were either too trivial or too difficult. Average scan attempts per clue ranged from 3.3 to 3.4, further confirming consistency across clues. The fact that timing variation was not driven by clue wording difficulty (since all four clues showed similar times) suggests the dominant source of session-to-session variance was environmental factors such as library crowding and student reading speed.

4.6.2 Gemini Vision Rejection Rate

Figure 4.9 illustrates the outcome distribution of Gemini Vision arrival confirmation scans. Of 199 total scan attempts across all fifteen sessions, 139 were rejected (69.8% rejection rate) and 60 were confirmed. Rejection rates per clue ranged from 68.1% to 71.4%, again showing consistency across locations.

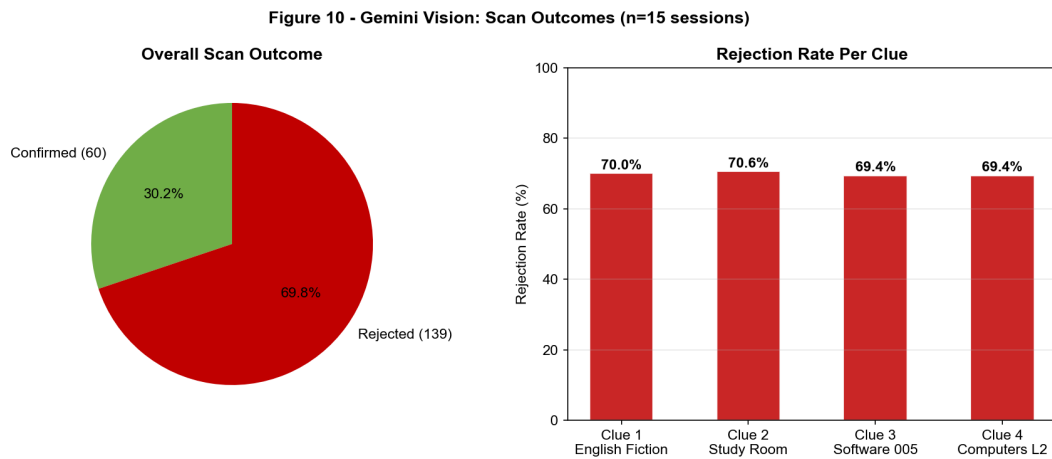


Figure 4.9: Gemini Vision scan outcomes: overall (left) and per-clue rejection rate (right) across 15 sessions.

A rejection rate of 69.8% is high and constitutes the primary technical limitation of the system. The confirmation prompt instructed Gemini to return a plain “YES” or “NO” response based on whether the photo contained visual evidence of the described shelf or location. In practice, variable lighting conditions, camera angle, and image blur caused conservative rejections. Crucially, however, every participant received at least one confirmation for every clue, meaning the hunt was always completable the rejection rate introduced friction rather than failure. This distinction is important: the system’s reliability (100% completion) remained acceptable despite its precision being sub-optimal.

Participants appeared to adapt to the scan behaviour quickly. Qualitative re-

sponses noted the need to “hold the phone steady” and “get closer to the shelf label,” indicating that participants learned effective scanning strategies within the session. Future work using a fine-tuned vision model or clearer in-app guidance for photo framing could substantially reduce the rejection rate.

4.7 Overall App Rating and Discussion

The overall application rating (Figure 4.10) produced a mean of 4.25 out of 5. Eight of twelve participants (67%) rated the application 4 or 5, one participant rated all items at 3 throughout the survey, and no participant awarded a rating below 3.

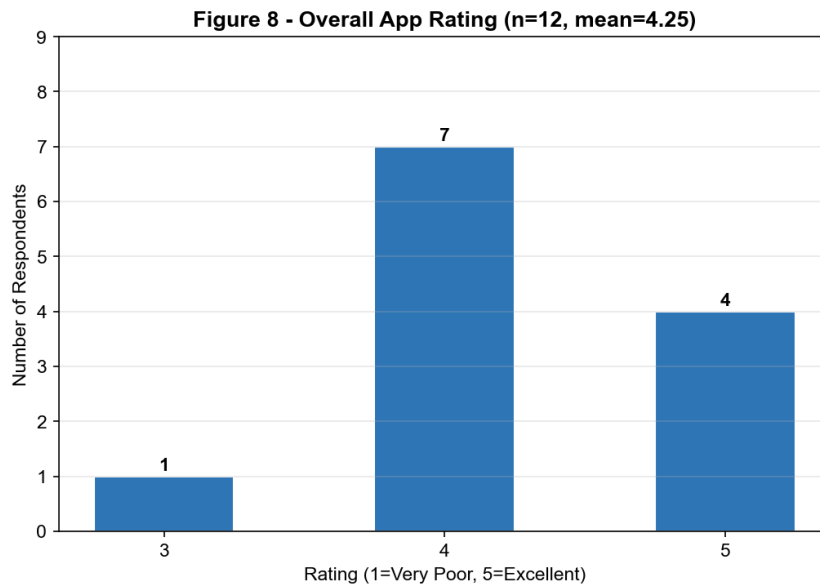


Figure 4.10: Overall application rating distribution (n=12, mean=4.25).

The convergence of results across all three research questions presents a coherent picture. For RQ1, both modes exceeded a section mean of 4.20, with recommendation intent reaching 4.58. For RQ2, LLM-generated clues were perceived as fun and sufficiently informative despite occasional re-reading require-

ments. For RQ3, the +1.50 confidence improvement and unanimous preparedness response provide the strongest evidence that the application achieves its primary orientation objective.

These findings are consistent with Akçayır and Akçayır's [7] systematic review, which identified direct physical engagement and immediate feedback as the key advantages of AR over passive learning tools. They also align with Kolb's [6] experiential learning model: the hunt engaged the concrete experience and reflective observation stages by requiring students to physically navigate, observe, and photograph library locations, creating richer memory encoding than reading a printed floor plan would achieve.

The integration of a large language model for clue generation demonstrated that AI-assisted content production is viable within a resource-constrained educational tool. Camilleri and Camilleri [11] argued that LLMs can reduce the content creation burden for educational designers while maintaining quality. This evaluation provides empirical support: the Gemini API generated contextually appropriate, engaging riddle clues from structured shelf data without any manual curation, and participants responded positively to the output quality.

4.8 Summary

This chapter presented quantitative and qualitative evidence from twelve survey respondents and fifteen application sessions. All three research questions received positive support from the data. RQ1 was addressed by section means above 4.20 for both application modes. RQ2 was addressed by a clue comprehension mean

of 4.19 and qualitative confirmation that LLM-generated clues were engaging and understandable. RQ3 was addressed by a +1.50 mean confidence improvement and unanimous positive preparedness response. The primary technical limitation identified was the 69.8% Gemini Vision rejection rate, which increased task effort without preventing task completion. These findings are discussed further in terms of limitations and future directions in Chapter 5.

Chapter 5: Conclusions and Recommendations

5.1 Summary of the Project

This dissertation set out to design, develop, and evaluate an Augmented Reality Android application the AR Library Navigator intended to address the well-documented limitations of traditional library orientation methods at MCAST. Traditional guided tours and static presentations have consistently been shown to produce limited long-term improvements in student navigation confidence or independent resource usage [1,2]. The motivating premise of this project was that a gamified, AI-assisted mobile application could provide a more engaging, memorable, and scalable alternative.

The application was developed over a five-month period from January to May 2026 using Unity 2022.3.15f1 LTS, the Vuforia Engine SDK, and the Google Gemini API. It comprises two primary student-facing modes. The **Treasure Hunt** mode presents students with a sequence of AI-reworded riddle clues, each corresponding to a real physical location within the MCAST library. Students navigate to each location independently, photograph it using the device's native camera, and receive an AI-powered YES/NO confirmation of their arrival via the Gemini Vision API. Progression through the clue sequence encourages students to physically explore areas of the library they might not otherwise encounter during a standard orientation. The **Book Search** mode provides a complementary utility, allowing students to select a topic from a dynamically populated drop-

down and receive plain-text floor and shelf directions based on the application's internal navigation graph. Together, the two modes address both the exploratory and the practical dimensions of library navigation.

Development followed an iterative, prototype-driven process. Several significant technical challenges were encountered and resolved, including conflicts between AR framework camera access and native photo capture, compressed texture encoding failures that silently prevented reference images from reaching the Gemini API, and race conditions in asynchronous response handlers that could freeze the application indefinitely. Each issue was diagnosed through on-device log analysis and resolved before the next test cycle, resulting in a stable, deployable application that was successfully tested with student participants at the MCAST library in May 2026.

The project also produced a 39-shelf library database of 1,160 book entries, a structured navigation graph covering both library floors, and a 34-question post-test survey designed to evaluate the application against three research questions.

5.2 Evaluation Against Research Questions

RQ1: Does an AR-based treasure hunt increase student engagement and perceived usefulness during library orientation compared to the existing tour format?

The evaluation provides clear positive evidence for RQ1. The Treasure Hunt mode produced a section mean of 4.32 out of 5 across five Likert items, with the highest score on *Would Recommend* ($M = 4.58$) and strong scores on *Aware of resources* ($M = 4.50$). The Book Search mode produced a section mean of 4.20,

with *Would Use Again* scoring highest at 4.45. No participant rated any item in either section below 3, and the overall application rating produced a mean of 4.25. Qualitative responses described the experience as more engaging and memorable than a standard tour, with participants noting that they physically explored areas of the library they had not previously visited. Taken together, these results support the conclusion that the AR-based treasure hunt did increase student engagement and perceived usefulness compared to the existing tour format, directly addressing RQ1 in the affirmative.

RQ2: To what extent do LLM-generated rewordings of orientation clues affect exploration behaviour, task completion time, and clue comprehension?

The clue comprehension section produced a mean of 4.19 out of 5. *Clues Felt Fun* scored highest ($M = 4.50$), confirming that the LLM-generated riddle format achieved an engaging tone. *Understood Without Help* scored lowest ($M = 3.83$), with three participants noting they re-read clues before identifying the target location indicating that the riddle format introduced productive interpretive challenge without causing failure. Session log data supports this: all fifteen sessions reached the `TREASURE_HUNT_COMPLETE` event, representing a 100% completion rate, and average time per clue was consistent across all four clues at 4.7 to 4.9 minutes, indicating balanced difficulty. The Gemini Vision rejection rate of 69.8% introduced scan friction but did not prevent task completion in any session. Together, these findings suggest that LLM-generated clue rewordings produced clues that were engaging, comprehensible, and consistent in their effect on exploration behaviour and task completion time.

RQ3: How do students describe the current library tour experience, and which specific aspects would they like improved?

Survey questions Q20–Q23 revealed that five of twelve participants (42%) had attended a previous library orientation but could not remember it well, and three participants (25%) had never attended any orientation. Pre-session navigation confidence averaged 2.42 out of 5 below the neutral midpoint indicating that prior orientations had not produced adequate independent navigation confidence in this cohort. When asked how they would find a book without the application, five of eleven participants stated they would browse shelves randomly, four would ask a librarian, and two would use a systematic approach, confirming that the existing tour format had not equipped participants with reliable independent retrieval strategies. Qualitative responses described the current tour as easy to forget and insufficiently hands-on. The aspects participants most commonly identified as needing improvement were: the ability to explore the library at their own pace rather than following a group, clearer guidance on how to locate specific books by subject, and a more interactive and memorable format. These findings directly characterise the limitations of the existing tour and establish the motivating gap that the AR Library Navigator was designed to address.

5.3 Limitations

Several limitations of this study should be acknowledged when interpreting the findings.

Sample size. Data was collected from a relatively small number of partic-

ipants. While this is consistent with the scope of an undergraduate final-year dissertation, the sample size limits the extent to which findings can be generalised to the broader student population. Quantitative results should be interpreted as indicative rather than definitive, and the qualitative responses are particularly valuable in providing contextual depth that compensates for the limited numerical sample.

Single institution. The application was designed and tested exclusively within the MCAST library environment. The navigation graph, shelf database, and clue locations are all specific to this institution, meaning that findings regarding usability and navigation effectiveness may not transfer directly to libraries with different layouts, cataloguing systems, or student demographics.

Absence of a control group. The study does not include a formal comparison group of students who received a traditional library orientation. While survey questions ask participants to reflect on previous orientation experiences, the absence of a controlled comparison condition limits the strength of conclusions regarding the relative effectiveness of the application versus existing methods.

Camera-based rather than AR overlay confirmation. Vuforia Engine was integrated during development and evaluated as a mechanism for marker-based location confirmation. In the final implementation, the native Android camera interface was used instead, due to hardware conflicts between the AR framework and the camera pipeline. As a result, the application does not provide augmented reality overlays in the traditional sense. While the camera-based photo confirmation is effective for arrival detection, it does not deliver the spatially embedded dig-

ital content that characterises full AR experiences, which may limit engagement compared to systems with richer AR overlays.

Limited treasure hunt coverage. The treasure hunt comprises four clue locations. While these cover both floors and several distinct library areas, they represent a small subset of the library’s full geography. Certain areas of the library could not be included in the clue sequence due to access restrictions most notably the dissertation room, which requires a formal permission request and permits only one student at a time, making it unsuitable for an unsupervised, self-paced orientation activity. Students who complete the hunt will therefore have explored a meaningful but incomplete portion of the space, which may limit the breadth of spatial learning achieved in a single session.

Gemini API dependency. Both clue rephrasing and arrival confirmation depend on the availability and responsiveness of the Gemini API. Network latency, API rate limits, and potential service interruptions introduce points of failure that are outside the application’s control. During testing, response times for multi-image vision requests ranged from approximately five to ten seconds, which introduces noticeable waiting periods in the treasure hunt flow.

Manual current location input in Book Search. The Book Search mode requires students to manually enter their current shelf number to receive floor-comparison directions. Students who do not know which shelf they are standing at cannot benefit from the directional guidance, reducing the feature’s effectiveness for users who are completely unfamiliar with the library layout.

5.4 Recommendations for Future Work

Based on the findings of this project and the limitations identified above, several directions for future development are proposed.

Automated indoor positioning. The most impactful enhancement to the Book Search mode would be the addition of automated current location detection, removing the requirement for manual shelf number input. This could be achieved through Bluetooth Low Energy (BLE) beacon networks, QR code scanning at shelf end-caps, or, if the Vuforia integration is extended, image target recognition at key library locations. Any of these approaches would allow the application to determine the student's current position automatically and provide directions without requiring prior spatial knowledge.

Full AR overlay navigation. Reinstating and extending the Vuforia AR camera component to deliver directional overlays such as arrows or highlighted paths rendered over the camera feed would bring the application closer to the AR-informed design originally envisioned and align it more fully with the literature on spatially embedded digital content [7, 8]. This would be particularly valuable for the Treasure Hunt mode, where students currently navigate between clues without in-app directional guidance.

Expanded clue set and library coverage. Extending the treasure hunt to ten or more locations, covering all major library sections including study rooms, computer areas, the borrowing desk, and all shelf sections, would provide more comprehensive spatial learning in a single session. The variant system already implemented in TreasureHuntRoute supports multiple path configurations, meaning

additional clues could be used to generate varied routes across different sessions.

Longitudinal retention study. The current evaluation captures student perceptions immediately after use. A follow-up study conducted several weeks after orientation would provide evidence on whether the spatial knowledge gained through the treasure hunt is retained over time, addressing a key weakness identified in the literature on traditional orientation methods [2].

Comparison with traditional orientation. A formal comparative study, in which one cohort receives the AR Library Navigator orientation and a control cohort receives the standard guided tour, would provide stronger evidence for or against the application's effectiveness relative to existing practice. This would address the absence of a control group identified as a limitation of the current study.

Multi-language support. Given MCAST's diverse student population, extending the application to support both Maltese and English interfaces including LLM-generated clues in both languages would improve accessibility and inclusivity.

Staff analytics dashboard. A companion web interface allowing library staff to view aggregated session log data including popular search topics, average hunt completion times, and clue locations where students frequently failed scans would support ongoing improvement of both the application and the physical library signage.

5.5 Concluding Remarks

The AR Library Navigator demonstrates that a gamified, AI-assisted mobile application can provide a technically viable and student-engaging alternative to tra-

ditional library orientation. By combining dynamic LLM-generated clue rephrasing with camera-based AI arrival confirmation and a practical book search utility, the application addresses both the motivational and the navigational dimensions of library orientation in a single deployable tool. The results of this evaluation are encouraging: all twelve participants completed the treasure hunt, all reported feeling more prepared to use the library independently, and mean navigation confidence improved by 1.50 points after a single session.

While the limitations of sample size, single-institution testing, and the absence of a control group caution against strong generalisable claims, the results of this study provide a meaningful foundation for further research and development. The application is not presented as a finished product but as a validated prototype that demonstrates the feasibility of the hybrid AR-AI orientation model and identifies clear directions for its extension.

Ultimately, the project contributes a working prototype, a structured evaluation framework, and a set of concrete recommendations that together advance the case for technology-enhanced library orientation as a scalable, student-centred alternative to the guided tour. The positive reception from student participants, many of whom had no prior experience with AR applications, suggests that the barrier to adoption is lower than might be assumed and that the primary requirement for wider deployment is refinement of the scanning mechanism and expansion of the clue and shelf coverage to fully represent the library's geography.

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Chapter A: Survey Instrument

The following reproduces the complete 34-question post-test survey administered to participants via Microsoft Forms immediately following their AR Library Navigator session. An asterisk (*) denotes a required question. Likert-scale items used a five-point star rating in the digital form.

Section A: Background

1. Is this your first time visiting this library? *
 - Yes
 - No
2. Before today, how confident were you in finding resources in this library?
*
(1 = Not confident at all, 5 = Very confident)
Scale: 1 2 3 4 5
3. Have you used an Augmented Reality (AR) app before? *
 - Yes
 - No
 - Not sure

Section B: Engagement and Usefulness

4. I found the AR treasure hunt engaging and enjoyable. *
(1 = Strongly Disagree, 5 = Strongly Agree)
Scale: 1 2 3 4 5
5. Using the app made me feel more confident about finding resources independently. *

(1 = Strongly Disagree, 5 = Strongly Agree)

Scale: 1 2 3 4 5

6. The treasure hunt motivated me to explore the library more than I normally would. *

(1 = Strongly Disagree, 5 = Strongly Agree)

Scale: 1 2 3 4 5

7. I would recommend this app to other students during orientation. *

(1 = Strongly Disagree, 5 = Strongly Agree)

Scale: 1 2 3 4 5

8. I felt more aware of where key library resources are located after completing the activity. *

(1 = Strongly Disagree, 5 = Strongly Agree)

Scale: 1 2 3 4 5

9. Compared to a standard guided tour, I think this app is a more effective way to learn the library layout. *

(1 = Strongly Disagree, 5 = Strongly Agree)

Scale: 1 2 3 4 5

10. What did you enjoy most about the AR treasure hunt experience? *

[Open text response]

11. Did anything frustrate or confuse you during the activity? Please describe.

If nothing, write "None". *

[Open text response]

12. If you have attended a standard library tour before, how does this app compare to that experience?

[Open text response]

Section C: Clue Comprehension

13. The clues were written in a way that was easy to understand. *

(1 = Strongly Disagree, 5 = Strongly Agree)

Scale: 1 2 3 4 5

14. The wording of the clues made the activity feel fun rather than like a test.

*

(1 = Strongly Disagree, 5 = Strongly Agree)

Scale: 1 2 3 4 5

15. I understood what each clue was asking me to find without needing help.

*

(1 = Strongly Disagree, 5 = Strongly Agree)

Scale: 1 2 3 4 5

16. The clue descriptions gave me enough information to locate the target. *

(1 = Strongly Disagree, 5 = Strongly Agree)

Scale: 1 2 3 4 5

17. Was there any clue that was particularly unclear or confusing? If yes, which one and why? If none, write "None". *

[Open text response]

18. In your own words, what did you think the clues were asking you to do?

*

[Open text response]

19. Do you think the way the clues were written affected how long it took you to find each location? Please explain. *

[Open text response]

Section D: Previous Library Experience

20. Have you previously attended a library orientation or guided tour? *

- Yes
- No
- I attended one but do not remember it well

21. How useful did you find the previous library orientation?

(1 = Not useful at all, 5 = Extremely useful)

Scale: 1 2 3 4 5

22. After your previous orientation, how well did you remember where things were located in the library?

(1 = I remembered nothing, 5 = I remembered everything clearly)

Scale: 1 2 3 4 5

23. What did you dislike most about the standard library orientation format?

[Open text response]

Section E: Book Search Feature

24. I was able to find the topic I was looking for using the dropdown menu.

*

(1 = Strongly Disagree, 5 = Strongly Agree)

Scale: 1 2 3 4 5

25. The text directions provided were clear and easy to follow. *

(1 = Strongly Disagree, 5 = Strongly Agree)

Scale: 1 2 3 4 5

26. I was able to locate the target shelf using the directions given by the app.

*

(1 = Strongly Disagree, 5 = Strongly Agree)

Scale: 1 2 3 4 5

27. I would use the Book Search feature in the future to find books in the library. *

(1 = Strongly Disagree, 5 = Strongly Agree)

Scale: 1 2 3 4 5

28. How would you normally find a book in the library without this app? *

- Ask a librarian
- Library computer catalogue
- Search online
- Look around myself
- I wouldn't know where to start

29. Is there anything about the Book Search feature you would change or improve?

[Open text response]

Section F: Overall

30. Which areas of the library are you most uncertain about finding on your own? Select all that apply. *

- Specific book sections / shelves
- Study rooms
- Computers and printing
- Borrowing and returns desk
- Quiet zones
- None, I feel confident finding everything
- Other

31. What improvements would you most want in a future library orientation? *

[Open text response]

32. Overall, how would you rate your experience with the AR Library Navigator app? *

(1 = Very poor; 5 = Excellent)

Scale: 1 2 3 4 5

33. After completing this activity, do you feel more prepared to use the library independently? *

- Yes, definitely
- Somewhat
- Not really
- No

34. Is there anything else you would like to share about your experience today?

[Open text response]

Chapter B: Library Database Sample

The AR Library Navigator Book Search feature is underpinned by a structured text database (`library_database.txt`) containing 1,160 topic entries across 39 shelves, exported from an SQLite source via a Python script. Each shelf entry records a shelf name, a unique numeric identifier, a floor number, and a list of subject topics with their Dewey Decimal Classification codes. Table B.1 presents a representative selection of eight shelves from the database.

Table B.1: Representative sample of eight shelves from the library database (1,160 entries, 39 shelves total).

Shelf Name	ID	Floor	Sample Topics
Accounting / General Management	17	2	Accounting [657], Bookkeeping [657.2], Cost accounting [657.42], Auditing [657.45], General management [658]
Business / Management / HR	15	2	Financial management [658.15], Personnel management [658.3], Recruitment [658.311], Executive management [658.4]
Chemistry / Engineering	14	2	Analytical chemistry [543], Organic chemistry [547], Technology [600], Electrical engineering [621.3]
Communications / Computer Engineering	22	2	Communications engineering [621.382], Computer engineering [621.39], Machine engineering [621.8]
Electrical Engineering / Electronics	21	2	Electrical machinery [621.31042], Wind power [621.312136], Lasers [621.366], Electronics [621.381]
Architecture / Building Materials	9	2	Architecture [720], Landscape design [712], Architectural drawing [720.284], City planning [711.4]
Disease / Engineering Materials	24	2	Diseases [616], Mental disorders [616.89], Engineering materials [620.11], Concrete [620.136]
Marketing / Industrial Technology	B1-05	1	Marketing channels [658.87], Telemarketing [658.872], Credit management [658.88], Advertising [659.11], Public

The full database file is included in the project repository and is loaded at runtime via `Resources.Load<TextAsset>`. The Book Search dropdown is populated dynamically from the shelf names at application start.

Chapter C: Key Code Listings

This appendix presents selected excerpts from three core scripts of the AR Library Navigator. Full source code is available at <https://github.com/lazytitan1234/ARLibraryNavigator>.

C.1 GeminiClassifier.cs Clue Rephrasing and Arrival Confirmation

`GeminiClassifier.cs` handles all communication with the Google Gemini API. The `RephraseClue` method sends a raw clue description to the text endpoint and returns an AI-generated riddle. The `PostVisionMultiImage` method sends reference photographs alongside the student's live camera frame and asks Gemini whether the same location is shown.

Listing C.1: GeminiClassifier.cs RephraseClue method

```
public void RephraseClue(string rawClueText,
                        Action<string> onSuccess,
                        Action<string> onFailure)
{
    string rephrasePrompt =
        "Rephrase the following library treasure hunt clue as a fun, " +
        "engaging riddle for students. Keep it concise (1-3 sentences). " +
        "Preserve the meaning the student must still find the same place. " +
        "Do not reveal the answer directly.\n\n" +
        "Original clue: \"" + rawClueText + "\"\n\n" +
        "Respond with ONLY the rephrased clue, nothing else.";

    StartCoroutine(PostRaw(rephrasePrompt, onSuccess, onFailure,
                           maxOutputTokens: 150, temperature: 0.7f));
}
```

Listing C.2: GeminiClassifier.cs multi-image arrival confirmation prompt

```
string promptText =
    "You are helping verify whether a student has found the correct " +
    "target in a treasure hunt.\n" +
    "The first " + refCount + " image(s) are REFERENCE photos of the target.\n" +
    "The last image is a LIVE photo taken by the student right now.\n\n" +
    "Does the live photo show the same target as the reference photos?\n" +
    "Be generous: if the main subject clearly matches, answer YES.\n\n" +
    "Reply with ONLY one word: YES or NO.";
```

C.2 SessionLogger.cs CSV Event Logging

`SessionLogger.cs` records all application events to a timestamped CSV file stored on the device. Every row contains an ISO 8601 timestamp, an 8-character

session identifier, an event type, and an optional data field. The file is flushed after every write to prevent data loss if the application terminates unexpectedly.

Listing C.3: SessionLogger.cs LogEvent and OpenLogFile methods

```
public void LogEvent(string eventType, string data = "")
{
    string timestamp = DateTime.UtcNow.ToString("yyyy-MM-ddTHH:mm:ss.fffZ");
    string row = timestamp + "," + _sessionID + "," +
        eventType + "," + EscapeCSV(data);

    if (_isOpen && _writer != null)
    {
        _writer.WriteLine(row);
        _writer.Flush(); // flush immediately so data survives a crash
    }
}

private void OpenLogFile()
{
    string fileName = logFilePrefix + "_" + _sessionID + "_" +
        DateTime.UtcNow.ToString("yyyyMMdd_HH:mm:ss") + ".csv";
    string filePath = Path.Combine(Application.persistentDataPath, fileName);
    _writer = new StreamWriter(filePath, append: false,
        System.Text.Encoding.UTF8);
    _writer.WriteLine("timestamp,session_id,event_type,data");
    _writer.Flush();
    _isOpen = true;
}
```

C.3 TreasureHuntController.cs Clue Display and Scan Flow

TreasureHuntController.cs manages the full Treasure Hunt user journey.

The ShowCurrentClueCoroutine method requests a rephrased clue from GeminiClassifier and enables the Scan Me button once a response is received. The scan flow then submits the student's photograph to Gemini Vision and routes the result to either clue progression or a rejection message.

Listing C.4: TreasureHuntController.cs clue display and Gemini Vision scan flow

```
private IEnumerator ShowCurrentClueCoroutine()
{
    var clue = CurrentClue();
    progressLabel.text = "Clue " + (_currentClueIndex + 1) +
        " of " + _shuffledClues.Count;
    clueText.text = "Preparing your clue...";
    scanMeButton.interactable = false;

    bool responseReceived = false;
    string displayText = clue.rawClueText;

    geminiClassifier.RephraseClue(clue.rawClueText,
        rewoded => { displayText = rewoded; responseReceived = true; },
        err => { responseReceived = true; });

    yield return new WaitUntil(() => responseReceived ||
        Time.unscaledTime > Time.unscaledTime + 20f);
}
```

```
        clueText.text = displayText;
        scanMeButton.interactable = true;
        sessionLogger?.LogEvent("CLUE_SHOWN", "index=" + _currentClueIndex);
    }

    // After user confirms photo, send to Gemini Vision for arrival check
    geminiClassifier.CheckArrival(
        clue.referenceImages, clue.visionTargetDescription, _pendingJpeg,
        () => { arrived = true; responseReceived = true; },
        err => { errorMsg = err; responseReceived = true; });

    yield return new WaitUntil(() => responseReceived ||
        Time.unscaledTime > scanDeadline);

    if (arrived)
    {
        sessionLogger?.LogEvent("CLUE_CONFIRMED", "index=" + _currentClueIndex);
        OnArrived();
    }
    else
    {
        sessionLogger?.LogEvent("SCAN_REJECTED", "reason=\"" + errorMsg + "\"");
        scanMeButton.interactable = true;
    }
}
```